

Riemann Stereographic Energy Coupling Reveals Motor-Imagery-Related Changes in Sensor-Space EEG Shared-Energy Density

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Abstract— Riemann Stereographic Energy Coupling (RSEC) detects simultaneous shared-energy events across EEG channels through analytic-signal construction, epoch-wise normalization, stereographic energy mapping, and fixed-precision matching. Evaluation used 106 participants from the PhysioNet EEG Motor Movement/Imagery dataset to compare rest, motor imagery, and motor execution. Broadband global RSEC density was lower during imagery than rest (paired $d_z = -0.354$, Holm-adjusted $p = 0.00444$) and execution ($d_z = -0.569$), whereas execution did not differ significantly from rest. Real imagery-minus-rest changes also differed from phase-randomized surrogate changes. Matched comparisons with band power, coherence, phase-locking value, phase-lag index, and weighted phase-lag index yielded imagery-related effect-size differences whose multiplicity-adjusted intervals all included zero. Frontopolar participation decreased in 78 of 106 participants. However, strict amplitude-based rejection using frontopolar channels reversed the mean global imagery–rest density difference in small retained subsets. RSEC provides a group-level descriptor of condition-related sensor-space EEG organization. The neural origin of the effects and their value for individual decoding remain unresolved.

Keywords— EEG, motor imagery, stereographic projection, energy coupling

I. INTRODUCTION

Electroencephalography (EEG) supports biomedical signal analysis through noninvasive recordings that resolve rapid changes in brain activity. The temporal resolution is especially important in motor imagery and brain–computer interface (BCI) studies, where task-related modulation of sensorimotor rhythms can occur within short time intervals [1, 2]. Despite this advantage, network-level EEG interpretation remains difficult. Scalp channels record mixtures of neural and non-neural sources, and apparent sensor-level coupling can be shaped by common-reference effects, volume conduction, and ongoing rhythmic activity [3–5]. Task-sensitive sensor-level summaries are therefore useful when interpreted separately from anatomical connectivity.

Motor imagery provides a useful setting for evaluating such features. Imagined movement recruits sensorimotor processes without overt movement and remains central to BCI and neurorehabilitation applications [2, 6]. Mo-

tor imagery and motor execution are related but not identical. Execution includes overt motor output and proprioceptive or sensory feedback, whereas imagery depends more strongly on internally generated motor planning with suppressed action. A feature that responds differently to imagery and execution may therefore capture motor-state-specific sensor-space changes rather than a generic rest-to-task change.

Network neuroscience commonly uses graph-theoretic approaches to study functional integration, segregation, modularity, and hub organization [7–9]. EEG network studies often begin with pairwise measures such as coherence, phase-locking value (PLV), phase-lag index (PLI), weighted PLI, or imaginary coherence [4, 5, 10, 11]. These measures are valuable, but each emphasizes a particular form of association. This paper evaluates a geometry-inspired alternative based on analytic-signal energy, Riemann stereographic projection, and fixed-precision shared-energy matching.

The proposed method, Riemann Stereographic Energy Coupling (RSEC), builds on earlier work on antipodal connectivity structures and Antipodal Hub Nodes (AHNs), where EEG signals were transformed into a complex-domain representation and analyzed through projection- and Riemann-based normalized transformations [12, 13]. Matching stereographic energy values are treated as transient sensor-space shared-energy events. Evaluation on the PhysioNet EEG Motor Movement/Imagery dataset [14–16] addresses three questions: whether density distinguishes imagery from rest and execution; how RSEC compares with conventional features; and how precision, preprocessing, and participant variability constrain interpretation.

II. RELATED WORK

During motor imagery, a movement is mentally rehearsed without being physically performed. EEG studies have consistently shown that imagined movement modulates ongoing sensorimotor rhythms, particularly through event-related desynchronization and synchronization in the alpha/mu and beta frequency bands [1, 2]. Neuper et al. further showed that imagery mode can influence the spatial distribution of EEG activity, with kinesthetic motor imagery producing activity patterns localized near sensorimotor hand areas [6]. These

findings establish sensorimotor rhythms and spatial response patterns as relevant context for evaluating motor-imagery features.

Functional connectivity analysis provides a framework for quantifying statistical dependencies among signals recorded from different EEG sensors or brain regions. Common EEG pairwise measures include coherence, PLV, imaginary coherence, PLI, and wPLI [4, 5, 10, 11]. PLV characterizes EEG synchronization by assessing how stable the relative phase of two signals remains over time or across repeated trials. However, phase-based connectivity measures, especially those sensitive to zero-lag synchrony, can be influenced by common sources, reference effects, and volume conduction. Imaginary coherence, PLI, and wPLI address part of this concern by emphasizing non-zero-lag interactions and reducing sensitivity to instantaneous coupling [4, 5, 11].

Although these measures are well established, they characterize particular forms of spectral or phase-based association and may not capture all task-sensitive structure in EEG signals. Nonlinear or geometry-inspired features may therefore provide complementary descriptions of sensor-space EEG dynamics. Direct experimental comparisons can assess their condition sensitivity relative to established measures.

Brain network hubs are typically defined using graph measures such as degree, strength, centrality, or participation across modules [8, 9]. In scalp EEG, sensors are not anatomical regions and edges depend on the selected coupling metric. RSEC builds on earlier AHN work [12, 13], but its shared-energy events represent sensor-level energy coincidences rather than graph-theoretic hubs. High channel participation indicates repeated involvement in these pairwise events.

Stereographic projection maps points from the extended complex plane onto the Riemann sphere and provides a geometric relationship between complex magnitude and chordal distance [17, 18]. This geometry underlies RSEC's energy representation of normalized analytic EEG signals. The earlier AHN framework emphasized antipodal connectivity and reported increased event counts during active engagement [12]. The present analysis examines motor-state dependence, including whether shared-energy density changes in the same direction during imagery and execution.

Conventional signal power provides a complementary benchmark because it summarizes squared amplitude without the RSEC transformation and matching steps. Together, ordinary power and conventional connectivity measures provide amplitude- and association-based comparisons for evaluating RSEC.

III. MATERIALS AND METHODS

III-A. Dataset and Preprocessing

The PhysioNet EEG Motor Movement/Imagery dataset was used for evaluation [14, 15]. Recordings were collected with the BCI2000 system [16]. The dataset contains 109 subjects with 64-channel EEG recordings during baseline, motor imagery, and motor execution tasks. The eyes-open baseline run R01 was used as the resting condition. Left/right motor imagery runs R04, R08, and R12 were used as the imagery condition, and left/right motor execution runs R03, R07, and R11 were used as the execution condition. Left and right task annotations were pooled within each condition; the analysis concerns motor state rather than left-versus-right decoding.

Subjects 88, 92, and 100 have 160-Hz baseline but 128-Hz task recordings; their exclusion defined the primary native-160-Hz cohort of 106. Cohort sensitivity analyses used continuous-signal and event resampling to include all 109 at common rates of 160 and 128 Hz. Subject 89, whose baseline annotation was atypical, was retained in the primary cohort and examined separately through exclusion checks. The five configurations were the primary 106-participant cohort at 160 Hz, that cohort without subject 89 ($n = 105$), all 109 at 160 Hz, all 109 without subject 89 at 160 Hz ($n = 108$), and all 109 at 128 Hz. This separates participant inclusion from sensitivity to the chosen common sampling rate.

All recordings were loaded using MNE-Python 1.7.1 [19]. EEG channel names were standardized and the standard 1005 montage assigned. Continuous runs were filtered separately with MNE's zero-phase FIR filter from 1 to 31 Hz. The baseline recording was divided into successive four-second intervals, with no samples shared between intervals. Task epochs began at T1/T2 event annotations and remained within the annotated task. Four-second epochs contained 640 samples at 160 Hz and 512 at 128 Hz. The primary analysis used 15 baseline windows and all available task epochs, with no task-epoch cap. Subject-level condition means were computed before statistical testing, so each subject contributed one paired value per condition. The recorded reference was retained, without amplitude rejection or independent component analysis (ICA); alternative reference and amplitude settings were evaluated separately. All 763 source EDF files matched the published checksums.

III-B. RSEC Shared-Energy Feature Extraction

The RSEC feature-extraction workflow is summarized in Fig. 1. Each EEG epoch was transformed into a complex analytic representation, mapped to a Riemann stereographic energy series, and summarized using shared-energy events across channel pairs.

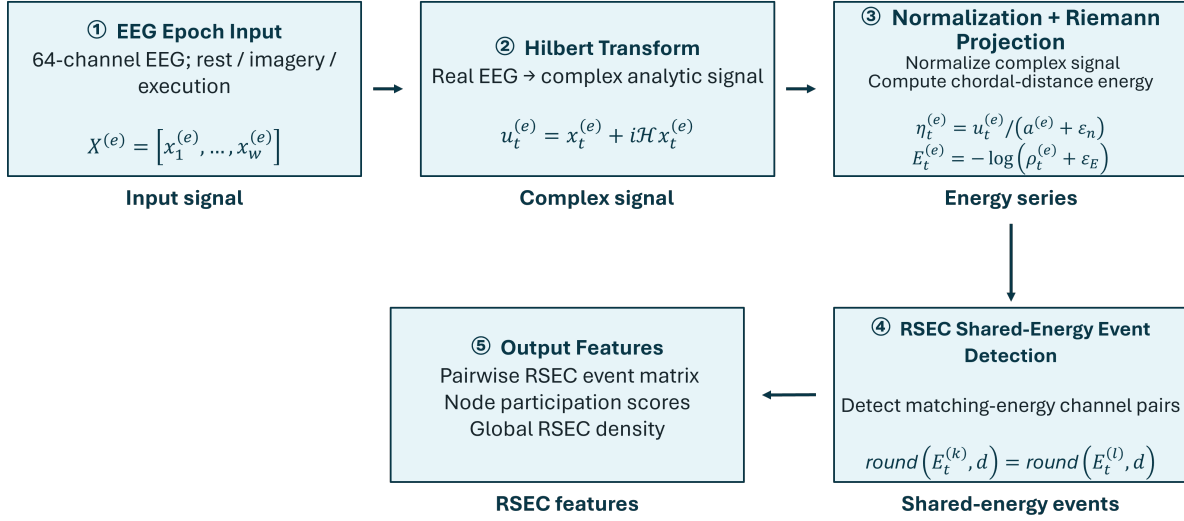


Figure 1. Overview of the RSEC feature-extraction pipeline. Each multichannel EEG epoch is transformed into a complex analytic representation using the Hilbert transform. Channel-wise analytic signals are normalized within each epoch and mapped through the Riemann stereographic framework to obtain chordal-distance energy. Shared-energy events are detected between channel pairs and summarized as pairwise event matrices, node participation scores, and global RSEC density.

Let $x_e(t)$ denote the EEG signal from channel e at time t within an epoch. The analytic signal is computed as

$$u_e(t) = x_e(t) + i\mathcal{H}\{x_e(t)\}, \quad (1)$$

where $\mathcal{H}\{\cdot\}$ is the Hilbert transform. For each channel and epoch, the analytic signal is normalized as

$$\eta_e(t) = \frac{u_e(t)}{a_e + \varepsilon_n}, \quad (2)$$

where $a_e = \max_t |u_e(t)|$ and ε_n is a small constant used to avoid division by zero. The normalized complex value is then mapped to a chordal-distance representation,

$$\rho_e(t) = \sqrt{\frac{1}{|\eta_e(t)|^2 + 1}}, \quad (3)$$

and the Riemann stereographic energy is computed as

$$E_e(t) = -\log(\rho_e(t) + \varepsilon_E). \quad (4)$$

Here, $\log$ denotes the natural logarithm.

For each time point, a pair of channels (k, l) is counted as an RSEC shared-energy event when

$$\text{round}(E_k(t), d) = \text{round}(E_l(t), d). \quad (5)$$

The implementation uses $\varepsilon_n = 10^{-12}$ with input in volts and dimensionless $\varepsilon_E = 10^{-12}$ for numerical stability. The primary precision is $d = 3$, corresponding to bins of width 0.001 in the transformed-energy coordinate. A shared-energy event occurs when two channels occupy the same transformed-energy bin at the same time. This is not a uniform pairwise-distance tolerance, because

nearby values can straddle a bin boundary. Global RSEC density is defined as

$$D = \frac{N_{\text{events}}}{TC(C-1)/2}, \quad (6)$$

where N_{events} is the total number of detected shared-energy events, T is the number of time samples in the epoch, and C is the number of EEG channels. Node participation scores were computed by counting how often each channel participated in shared-energy events and normalizing this count by the total number of possible events involving that channel.

The energy coordinate is dimensionless rather than physical energy in joules. The event rule compares transformed magnitudes at the same time point; it does not require matching phase angles. Node participation summarizes a sensor's coincidences with the electrode array rather than its local amplitude or cortical activation.

III-C. Conventional Comparisons

Five conventional features were evaluated independently: mean squared real-signal amplitude (band power, in μV^2), magnitude coherence, PLV, PLI, and wPLI. RSEC and all comparators shared participants, channels, reference, filters, and epoch indices. The comparisons evaluate the full RSEC pipeline and do not isolate the contributions of channel-wise normalization, stereographic mapping, and fixed-precision matching. Because the baseline supplied 15 epochs, the conventional comparison used all 15 rest epochs and 20 fixed-seed selections of 15 task epochs without replacement. Selection averages were computed within participant;

overlapping selections were not independent observations. The primary RSEC analysis used all available task epochs, whereas the conventional comparison used balanced 15-epoch samples.

MNE-Connectivity 0.7.0 estimated coherence, PLV, PLI, and wPLI across epochs using Fourier spectra with a Hann taper [20]. Scores were averaged over frequency bins and the 2,016 unordered channel pairs. Power was averaged over samples and channels without Hilbert transformation, RSEC normalization, or energy matching. Conventional estimators used their standard intrinsic normalization. Bands were broadband (1–31 Hz), mu (8–13 Hz), and beta (13–30 Hz); both narrow bands included 13 Hz. A spectral sensitivity check excluded the 1-Hz bin, using 1.25–31 Hz on the same broadband-filtered epochs to satisfy a five-cycle minimum in four-second windows.

III-D. Surrogate Control and Statistical Analysis

A phase-randomized surrogate control tested the imagery–rest contrast [21]. Rest and imagery epochs were independently phase-randomized, with one realization per channel and epoch preserving Fourier magnitudes. RSEC was recomputed with identical parameters for both surrogate conditions. The paired contrast was $\Delta D_{\text{real}} - \Delta D_{\text{surrogate}}$, where $\Delta D = D_{\text{imagery}} - D_{\text{rest}}$.

Participants were the inferential unit. Two-sided Wilcoxon signed-rank tests used the normal approximation, omitted zero differences, and applied no continuity correction. Signed Cohen’s d_z was the mean paired difference divided by its sample standard deviation. Mean-difference 95% percentile bootstrap intervals used 10,000 participant resamples; these estimate means and need not agree with rank-test significance. Four primary contrasts formed one Holm family: imagery–rest, execution–rest, imagery–execution changes, and real–surrogate imagery changes. The imagery–execution difference equals the difference of task-minus-rest changes because the same rest value is used.

The matched comparison formed a separate 54-test Holm family (six methods, three bands, three contrasts). Between-method differences were assessed using 100,000 paired-participant bootstrap resamples shared across methods for baseline-minus-RSEC differences in $|d_z|$. Bonferroni percentile intervals at 99.8889% per comparison targeted approximate 95% simultaneous coverage across 45 differences.

III-E. Sensitivity and Spatial Analysis

Seven alternative RSEC settings comprised $d = 2, 4, 5$, average reference, removal of Fp1/Fpz/Fp2, and mu and beta-band filtering. Their three contrasts formed a 21-test Holm family. Five cohort/rate configurations formed a separate 15-test family. Imagery node changes were tested across 64 channels with Holm

correction. Three regional tests compared frontopolar (Fp1/Fpz/Fp2) and sensorimotor (C3/Cz/C4) means with zero and with each other; participant direction counts assessed consistency.

Descriptive artifact-sensitivity summaries examined a spherical-spline surface Laplacian [22], median epoch aggregation, removal of the highest-amplitude 20% of epochs, and maximum peak-to-peak limits of 150, 250, and 500 μV over all or frontopolar channels. Thresholded analyses required at least five epochs per condition. For each retained cohort, results with and without amplitude-based epoch rejection were compared using the same 1–31-Hz preprocessing to distinguish epoch-selection effects from cohort composition.

IV. RESULTS

IV-A. Global Density and Participant Consistency

Motor imagery produced a significant reduction in global RSEC density relative to rest in the 106-subject cohort (rest: 0.020489, imagery: 0.018230; mean difference -0.002258 , raw $p = 0.00148$, $d_z = -0.354$). In contrast, motor execution did not produce a significant global change (mean 0.020487, raw $p = 0.3665$, $d_z = -0.0003$). The imagery–execution contrast was also significant (mean difference -0.002257 , raw $p = 8.02 \times 10^{-10}$, $d_z = -0.569$). Table 1 reports all primary contrasts, corrected probabilities, and confidence intervals.

The imagery–rest difference was negative in 66/106 participants (62.3%), and imagery–execution was negative in 84/106 (79.2%). Figure 2 shows the full participant distributions, including changes opposite to the group mean. These counts describe directional consistency; they are not estimates of classification accuracy. The near-zero execution mean also coexists with appreciable participant variability rather than uniformly absent changes.

IV-B. Surrogate and Cohort Checks

The phase-randomized surrogate analysis tested whether the imagery-related reduction was reproduced after disrupting temporal structure while retaining Fourier magnitudes. The surrogate imagery–rest mean was 0.000031 (raw $p = 0.108$), compared with -0.002258 in the real data. Real-minus-surrogate changes averaged -0.002289 (raw $p = 0.00153$, Holm $p = 0.00444$, $d_z = -0.353$). Thus, the surrogate control did not reproduce the real effect’s direction or magnitude. Figure 3 displays the paired participant differences, and Table 1 includes their uncertainty. The imagery–rest, imagery–execution, and real–surrogate primary contrasts survived four-test Holm correction.

Including all 109 participants at common 160 and 128 Hz yielded imagery–rest $d_z = -0.349$ and -0.348 ,

Table 1. Primary all-epoch RSEC contrasts ($n = 106$). Mean differences and their 95% participant-bootstrap intervals are scaled by 10^{-3} . p_{H4} adjusts the four primary tests; intervals are not multiplicity-adjusted. Negative counts refer to paired differences below zero.

Contrast	Mean	95% CI	d_z	p_{H4}	Negative
Imagery–rest	−2.258	[−3.489, −1.066]	−0.354	0.00444	66/106
Execution–rest	−0.002	[−1.226, 1.198]	−0.0003	0.36652	49/106
Imagery–execution	−2.257	[−3.020, −1.514]	−0.569	3.21×10^{-9}	84/106
Real–surrogate changes	−2.289	[−3.534, −1.079]	−0.353	0.00444	66/106

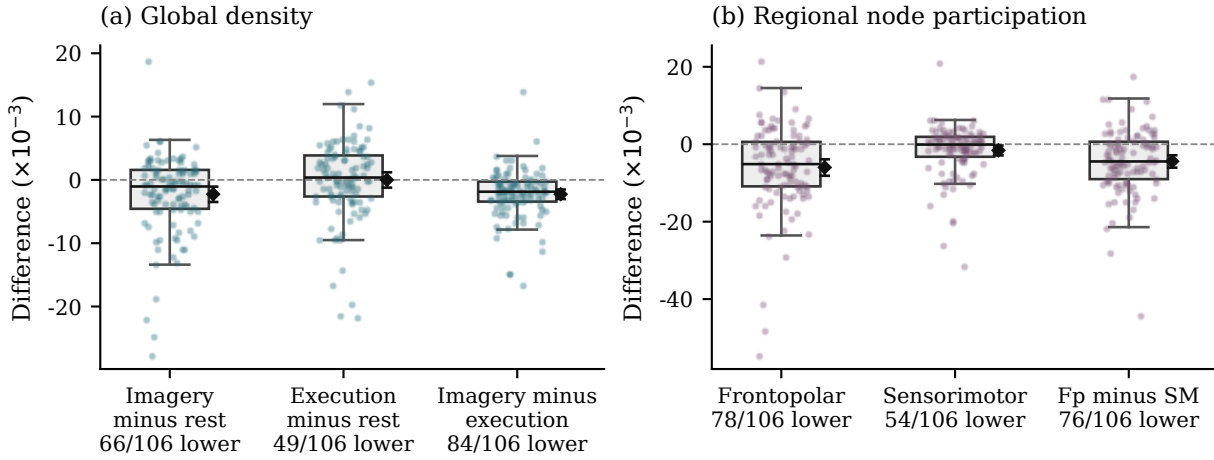


Figure 2. Participant-level RSEC differences ($n = 106$). (a) Global density contrasts. (b) Imagery–rest node participation over frontopolar (Fp: Fp1/Fpz/Fp2) and sensorimotor (SM: C3/Cz/C4) electrodes and their regional difference. Points are participants; boxes show median and interquartile range with 1.5-IQR whiskers. Diamonds and bars show means and 95% participant-bootstrap intervals. Labels count negative differences.

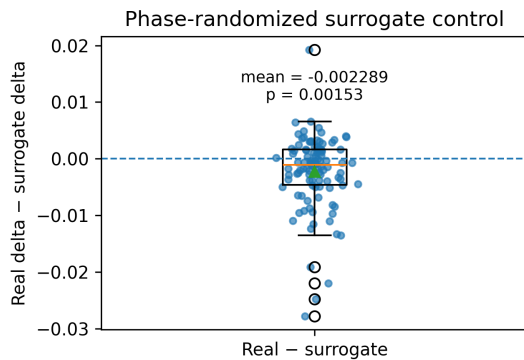


Figure 3. Phase-randomized surrogate control. Each point is one participant's real imagery–minus–rest change minus the surrogate change ($n = 106$). The green triangle marks the mean; the box shows the median and interquartile range, with 1.5-IQR whiskers. Open circles are boxplot outlier markers for participants also shown by filled points. The displayed $p = 0.00153$ is unadjusted; the four-test Holm value is 0.00444. Negative values indicate a more negative real effect.

and imagery–execution $d_z = -0.566$ and -0.571 . Both imagery contrasts survived the 15-test cohort correction in all five configurations; execution–rest did not. The persistence of these contrasts after including the three mixed-rate participants indicates that their exclusion was not responsible for the main imagery findings.

IV-C. Comparison with Conventional Features

Table 2 reports all 54 matched comparisons. Broadband imagery–rest d_z was -0.343 for RSEC, -0.084 for power, and -0.006 to $+0.271$ for connectivity measures. No broadband imagery–rest test survived the 54-test correction (RSEC adjusted $p = 0.0615$). This differs from the primary all-epoch result because epoch sampling and the multiplicity family differ. All six methods distinguished broadband imagery from execution after correction; RSEC had $d_z = -0.545$ and adjusted $p = 1.70 \times 10^{-7}$.

The direction and magnitude of imagery-related effects varied across bands and features (Table 2). Figure 4 shows the 30 imagery-related baseline–minus–RSEC effect-size differences from the 45-comparison family; all 30 intervals cross zero. For broadband execution–rest, the paired intervals supported larger effect magnitudes for coherence, PLV, and wPLI than for RSEC.

Table 2. Signed paired effect sizes for matched conventional comparisons ($n = 106$). I: imagery; E: execution; R: rest. Broadband: 1–31 Hz; mu: 8–13 Hz; beta: 13–30 Hz. A star marks Holm-adjusted $p < 0.05$ across all 54 tests, not superiority over another method. All methods use the same 15-epoch selections.

Band	Contrast	RSEC	Power	Coherence	PLV	PLI	wPLI
Broadband	I–R	−0.343	−0.084	+0.246	+0.271	−0.006	+0.207
Broadband	E–R	−0.0004	+0.335	+0.548*	+0.579*	+0.274*	+0.517*
Broadband	I–E	−0.545*	−0.496*	−0.402*	−0.470*	−0.433*	−0.365*
Mu	I–R	+0.208	+0.327	+0.498*	+0.547*	+0.210	+0.335*
Mu	E–R	+0.408*	+0.222	+0.620*	+0.652*	+0.209	+0.290
Mu	I–E	−0.298	+0.188	−0.208	−0.189	−0.061	+0.006
Beta	I–R	+0.186	+0.253	+0.184	+0.229	+0.059	+0.244
Beta	E–R	+0.459*	+0.150	+0.316*	+0.351*	+0.233	+0.433*
Beta	I–E	−0.419*	+0.234	−0.241*	−0.258*	−0.309*	−0.312*

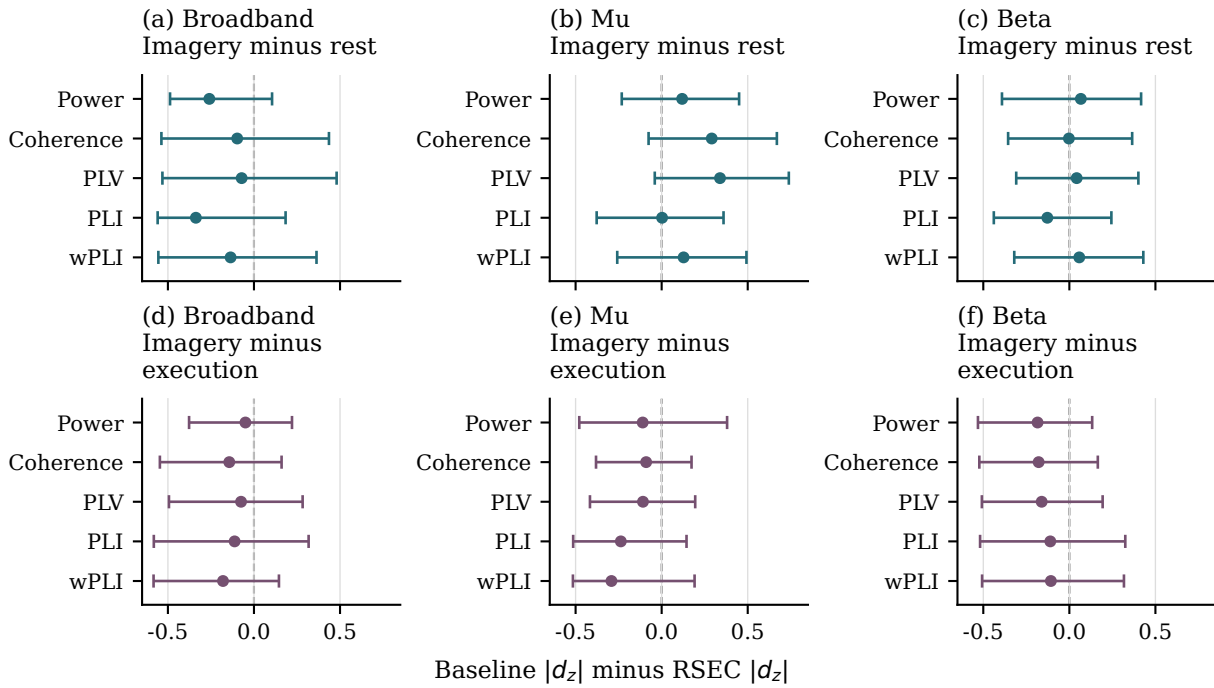


Figure 4. Comparative uncertainty for the six imagery-related settings ($n = 106$). Points show baseline $|d_z|$ minus RSEC $|d_z|$; negative values indicate a larger RSEC point effect. Bars are paired-participant percentile-bootstrap intervals at 99.8889% per comparison, targeting approximate 95% simultaneous coverage across all 45 comparisons, including execution–rest. All 30 imagery intervals cross zero. Signed effects appear in Table 2.

Excluding the 1-Hz spectral bin changed connectivity d_z by at most 0.0153 and preserved the ordering of broadband imagery point estimates.

Table 2 reports signed effect sizes to show the direction of change for each feature.

IV-D. Precision and Preprocessing Sensitivity

Table 3 shows precision and preprocessing dependence. Imagery–rest means remained negative at $d = 2, 4, 5$, but only $d = 4$ survived the 21-test adjustment. Average reference preserved a negative significant effect; removing frontopolar electrodes retained a negative mean

but not adjusted significance. Narrow-band imagery–rest effects were positive and nonsignificant after correction. Imagery–execution remained negative and significant in all seven all-epoch settings. The balanced mu imagery–execution comparison in Table 2 did not survive its separate 54-test correction.

IV-E. Spatial Findings and Artifact Sensitivity

Twenty-three of 64 channel-wise imagery changes survived Holm correction. Fpz, Fp2, and Fp1 decreased in 79, 75, and 74 participants, respectively. The frontopolar regional mean decreased in 78/106 participants ($d_z = -0.535$, Holm-adjusted $p = 2.06 \times 10^{-7}$), compared

Table 3. All-epoch RSEC sensitivity ($n = 106$). Entries are signed d_z ; stars denote Holm $p < 0.05$ across seven settings and three contrasts (21 tests). Only imagery-related contrasts are shown; the correction family also includes execution–rest tests, available in the supplementary results. The primary $d = 3$ result belongs to Table 1’s separate family.

Setting	I–R	I–E
$d = 2$	−0.268	−0.606*
$d = 4$	−0.384*	−0.544*
$d = 5$	−0.316	−0.472*
Average reference	−0.471*	−0.572*
Without Fp1/Fpz/Fp2	−0.314	−0.555*
Mu, 8–13 Hz	+0.208	−0.319*
Beta, 13–30 Hz	+0.194	−0.449*

with 54/106 for the sensorimotor region (Holm-adjusted $p = 0.203$). The frontopolar-minus-sensorimotor difference was negative in 76/106 (Holm-adjusted $p = 3.81 \times 10^{-7}$). These probabilities use the three-test regional family.

The imagery–rest global density difference depended on amplitude-based epoch rejection. Removing the largest-amplitude 20% retained all 106 participants and negative imagery–rest means. Frontopolar 150/250- μ V limits retained only 15/34 participants and reversed the means to +0.001501/ +0.001651, despite negative means without rejection in those same subsets. At 150 μ V, imagery–execution also approached zero ($d_z = +0.054$, $n = 15$). The strictest all-channel threshold retained only four participants. The surface Laplacian produced imagery–rest $d_z = -0.277$ and imagery–execution $d_z = -0.522$.

V. DISCUSSION

V-A. Motor-State-Dependent Shared-Energy Organization

RSEC density differed between motor imagery and execution in the primary broadband analysis. Imagery also reduced density relative to rest, whereas execution showed no significant global change. The imagery–execution contrast, its direction in 84 of 106 participants, and its persistence in the cohort/rate checks support group-level motor-state sensitivity rather than a uniform rest-to-task shift.

One possible interpretation is that imagery shifts the signal away from broadly shared resting-state energy patterns toward a more selective internally generated motor state. Execution combines motor commands with proprioceptive and sensory feedback, so regional changes could yield a different net global density. These are hypotheses about the observed feature, not demonstrated neural mechanisms. In particular, a near-zero execution–rest density change does not imply absent execution–

related EEG modulation: conventional coherence and phase measures responded strongly to execution. Global averaging can also conceal opposing pairwise or regional responses.

Increased event counts during active engagement in the earlier AHN work [12] and decreased RSEC density during motor imagery here concern different experimental conditions and event summaries. Shared-energy density therefore appears task-dependent rather than a universal index of engagement. The reversal of imagery–rest effect direction in the mu and beta analyses further illustrates its frequency dependence.

V-B. Relationship to Conventional Features

RSEC summarizes coincident transformed-energy bins, whereas ordinary power summarizes squared real-signal amplitude and conventional connectivity measures characterize spectral or phase relationships. Their differing response profiles reflect distinct feature definitions. The imagery-related paired intervals permit a range of relative effect magnitudes and establish neither superiority in either direction nor equivalence. Incremental value beyond conventional features would require a joint predictive evaluation rather than comparison of marginal effect sizes alone.

The primary analysis and matched comparison used different epoch selections and multiple-testing families. The primary analysis evaluated all-epoch RSEC contrasts within a four-test family, whereas the matched comparison evaluated six features across three bands and three contrasts. These differences account for the different adjusted significance decisions. Fixed global averaging may also discard useful spatial information for every method, and phase estimates based on 15 epochs remain subject to finite-sample uncertainty.

V-C. Interpretation of Frontal Effects

The strongest participation changes were concentrated over frontal and frontopolar sensors, with a frontopolar decrease in 78 of 106 participants. This is a recurring sensor-level pattern rather than a universal individual response. Its spatial distribution need not match local mu/beta power desynchronization: each participation score counts matches with the entire electrode array. A frontal sensor can therefore change through its relationships with other sensors without indicating local frontal cortical activation.

Motor planning, attention, and task engagement are possible neural contributors, but ocular activity, common sources, and reference effects are competing explanations [3, 4]. The negative imagery–rest means under average reference, after removal of frontopolar electrodes, and with the surface Laplacian show that the direction persists under several spatial preprocessing choices.

Strict frontopolar amplitude thresholds, however, reversed the global imagery–rest mean even relative to data without amplitude-based rejection from the same retained participants, indicating sensitivity to epoch selection in addition to cohort selection. Those thresholds left small cohorts, and the strictest frontopolar mask largely removed the imagery–execution effect. Amplitude rejection can remove genuine EEG activity as well as artifacts, so the contribution of artifacts remains unresolved. Electrooculography (EOG)-informed assessment and source-level validation would be needed to distinguish neural and non-neural explanations for the frontal pattern.

V-D. Scope and Limitations

The phase-randomized surrogate control did not reproduce the real imagery-related reduction, supporting sensitivity to the structure altered by phase randomization. Because this operation can change amplitude envelopes as well as temporal relationships, the result does not isolate phase coupling or prove causal interaction. One surrogate realization per epoch also limits assessment of variability across surrogate realizations.

The analysis concerns one public dataset, the left/right motor imagery and execution paradigm, and condition-level sensor-space summaries. Precision sensitivity characterizes dependence on fixed-precision matching but does not validate $d = 3$ as physiologically optimal. Sampling-rate checks reduce concern about exclusion of the three mixed-rate participants; matching published file checksums establishes data integrity, not independent verification of the acquisition timebase. No EOG-based adjudication or individualized source localization was performed.

Mean-difference bootstrap intervals and Wilcoxon tests target different aspects of the paired distribution. For example, the sensorimotor regional mean interval excludes zero, but its Wilcoxon test is nonsignificant. This distinction matters when changes are unevenly distributed across participants. The multiplicity-adjusted paired effect-size intervals are approximate bootstrap intervals, not distribution-free guarantees or equivalence tests.

RSEC is presently supported as a group-level descriptor and a candidate feature for subsequent BCI evaluation. A practical decoder would require held-out participants or sessions, training-only parameter selection, appropriate artifact handling, and direct assessment of incremental predictive value. Pairwise and spatial features may be useful candidates, but no classifier was evaluated in this study.

VI. CONCLUSION

RSEC identified group-level differences between motor imagery and both rest and execution in broadband sensor-space EEG. The imagery–execution contrast persisted across the examined cohort, sampling-rate, and precision configurations, whereas the imagery–rest effect depended more strongly on analysis settings. Conventional-feature comparisons left relative imagery-related effect magnitudes uncertain. Frontal participation decreases were common across participants, but their neural origin remains unresolved. Global density contrasts were sensitive to amplitude-based epoch rejection. Further evaluation should determine whether RSEC provides predictive information beyond conventional features under validated artifact handling and held-out testing.

DATA AND CODE AVAILABILITY

The EEG dataset is publicly available from PhysioNet [15]. The RSEC implementation is hosted at <https://github.com/bayazitkaraman/RSEC-motor-imagery>. Numerical summaries, participant-level figure inputs, and scripts that regenerate the reported tables and figures are included in the source archive, `SPMB_2026_latex_source.zip`, available in the repository's `docs/spmb_2026/` directory. Raw EEG recordings are not redistributed.

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